

Venkatesh Terikuti

Senior Data Scientist | Analytics Leader | Applied AI Researcher

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Professional Summary

Senior Data Scientist and ML Platform Engineer with 6+ years of experience designing and deploying scalable machine learning and analytics systems across cloud and enterprise environments. Specialized in building robust ML pipelines for large-scale data processing, anomaly detection, and predictive analytics. Hands-on with modern MLOps, distributed data architectures, and secure data management. Proven ability to drive product impact by bridging ML research, infrastructure, and cross-functional teams.

EXPERIENCE

ML Research Assistant (Post Grad research)

Feb 2025 – Present

University of Arizona

Tucson, AZ

- Built a scalable semantic search engine for e-commerce by fine-tuning Sentence Transformers and Google FLAN-T5 on a 1.3M-product catalog, integrating FAISS for fast vector retrieval and enabling real-time, intent-aware search.
- Implemented a full-stack ML pipeline covering data prep, model fine-tuning, and dynamic filter extraction (e.g., price, ratings), enhancing structured query refinement and delivering a 90% increase in Precision@10 over traditional keyword search.
- Led applied research on LLM-powered search, authored a peer-reviewed paper titled "*LLM-based Semantic Search for Conversational Queries in E-commerce*" submitted to CIKM'25 (ACM), showcasing innovative use of Sentence-BERT for user query understanding and product discovery for public sector use cases.

Senior Software Engineer – Data Scientist

Nov 2021 – Jan 2023

GlobalLogic

Remote

- Built deep learning pipelines using TensorFlow and attention-based LSTMs to forecast demand for 100K+ SKUs with 96% accuracy; integrated models via CI/CD using Jenkins and GitLab CI.
- Designed and deployed a centralized MLOps platform on AWS SageMaker, serving batch and real-time features for 10+ models; reduced model deployment latency by 40%.
- Engineered ETL workflows in MySQL, Hive, and Airflow; utilized Databricks for scalable data processing and collaborative analytics. Developed a real-time anomaly detection pipeline (Python, Kafka) to identify drops in order placement and app responsiveness, enabling proactive resolution of customer-impacting issues.
- Partnered with marketing and supply chain stakeholders to deliver insights and dashboards supporting product inventory and campaign effectiveness.
- Conducted A/B test evaluation and user segmentation to support growth experimentation strategies; delivered insights used to prioritize product roadmap items and optimize user journeys. Engaged directly with cross-functional teams in Agile/Scrum environments; influenced technical decisions and led process improvements.

Senior Data Analyst

Nov 2019 – Oct 2021

Geetham Software (Client: Lytx)

Remote

- Led a distributed analytics and DevOps team (US, Argentina, Poland) to deliver big data solutions supporting millions of telematics events daily.
- Built and managed SQL-based and Databricks Delta Lake pipelines for ingesting 6M+ daily events, ensuring 99.9% data freshness for real-time analytics, cleaning, and modeling large-scale event and customer data.
- Established dbt Core semantic models and reusable data marts for analytics self-service; drove advanced cohort and A/B testing analytics.
- Designed and deployed automated workflows in AWS Glue and Airflow for campaign, CRM, and compliance datasets; implemented Tableau dashboards, reducing report turnaround by 40%.
- Provided production troubleshooting, performance tuning, and issue resolution across global deployments. Partnered with product to design and analyze 10+ A/B tests (e.g., promotional offers), delivering actionable insights that improved conversion by 8%..
- Architected and implemented dimensional data models (star and snowflake schemas) in Snowflake and Databricks, supporting scalable, performant analytics and ad hoc reporting across sales, operations, and compliance functions.

Data Analyst

Mar 2016 – Oct 2019

Geetham Software

Chennai

- Developed ETL workflows for campaign, finance, and CRM data across AWS and Azure, automating ingestion and cleaning tasks.
- Conducted advanced segmentation, churn analysis, and predictive modeling using Python, SQL, and Tableau.
- Maintained and improved daily reporting pipelines, standardized data for consistent BI consumption, and supported quarterly business reviews.
- Acted as key point of contact for resolving pipeline issues and ensuring high data quality standards.
- Led analytics delivery in compliance-driven environments, establishing data governance protocols, ensuring data quality, and

partnering with legal and finance teams to enable secure, audit-ready reporting for regulatory and operational reviews.

EDUCATION

Master of Science in Data Science, The University of Arizona (GPA: 3.89)

Dec 2024

Bachelor of Technology in Mechanical Engineering, Acharya Nagarjuna University (GPA: 3.42)

May 2015

PROJECTS

Building LLM From Scratch.

- Implemented GPT-style large language model from ground up using PyTorch with byte-level tokenizer and scaled dot-product attention. Built complete training pipelines including pretraining on unlabeled text, classification finetuning, and instruction finetuning (SFT). Demonstrated deep understanding of attention mechanisms, gradient stabilization, and modern LLM architectures.

Transformer From Scratch: Scaling Journey.

- Built complete transformer implementation scaling from 2.4M to 52M parameters with explicit einsum operations. Compared MacBook character-level models vs H100 GPU BPE models, analyzing hardware efficiency and training dynamics. Implemented production infrastructure with mixed precision training, memory mapping, and automated benchmarking.

High-Performance Video Diffusion Model Optimization – Wan 2.2

- Optimized the Wan 2.2 text-to-video diffusion model pipeline for H100 GPUs, leveraging advanced context-parallelism, flash attention 3, and direct cuBLAS/triton kernel integration to achieve sub-25s 720p video generation for 81 frames—outperforming standard FA2/FA3 baselines (15–30 TFLOPS, 2–4× speedup).

Predicting Customer Churn in E-Commerce

- Created churn prediction models using Random Forest, SVM, and XGBoost with >90% accuracy on the 'Online Retail II' dataset (500K+ records).
- Conducted retention cohort analysis and lifetime value estimation; tuned model performance using cross-validation and grid search. Delivered Tableau dashboards and actionable insights to simulate real-world e-commerce churn scenarios and KPIs.

Enterprise Knowledge Copilot: Agentic RAG for Internal Q&A & SOP Automation

- Developed an enterprise-grade RAG platform for knowledge workers to ask natural language questions against internal documentation (SOPs, runbooks, tech specs, policy PDFs).
- Leveraged LangChain with pgVector and Redis to embed, retrieve, and rerank context chunks; OpenAI GPT-4 for synthesis and summarization. Designed multi-agent orchestration for use cases like IT support, onboarding, compliance checks, and change management documentation.

Skills

Languages: Python, SQL, Bash, C++

Big Data & Cloud: Spark, PySpark, Databricks, Airflow, AWS (Lambda, Glue, S3, EC2), Azure (Data Factory, Data Lake), Snowflake, Hadoop, Kubernetes, Delta Lake, dbt Core

Data Engineering: ETL/ELT Pipelines, Data Modeling, Data Cleaning/Standardization, Data Warehousing

BI & Visualization: Tableau, Power BI, Google Data Studio

DevOps: Docker, CI/CD, GitLab CI, Jenkins, Terraform (IaC), Prometheus/Grafana

Data & ML Platforms: SQL, dbt Core, Airflow, Tableau, Looker (familiar), Power BI, Databricks, Snowflake, AWS (Lambda, S3, Glue, SageMaker), Data Warehousing

Statistical Modeling: Regression, Classification, Clustering, Experimentation/A/B Testing, Causal Inference

Leadership: Agile/Scrum, Cross-functional Team Leadership, Mentorship, Analytics Enablement, Stakeholder Management, Change Management, OKR Development

Certifications

Deep Learning Specialization – Andrew Ng, Coursera

SQL(Advanced) - Hackerrank